

Two-Copy Parameter Twirling: A Worked Example

3-Qubit YZY Entangling Circuit

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Run: `ansatz=YZY_ENTANGLING`, $n = 3$, $L = 2$, $d = 1$, RX encoders, $O = \frac{1}{3} \sum_{i=0}^2 Z_i$
Code: https://github.com/quantumsoftwarelab/Circuit_Harmonic_Matrices

Abstract

The covariance of the Fourier coefficients of a data-encoded quantum model can be written as a contraction $\Sigma = CPC^\dagger$ of the (potentially exponentially large) circuit-harmonic matrix C . This note shows how the two-copy parameter-torus twirl evaluates that contraction *without explicitly building* C using Pauli back propagation and a worked example for a simple circuit. Working on a 3-qubit, 2-layer YZY entangling ansatz with RX data encoders and mean-magnetisation observable $O = \frac{1}{3}(ZII + IZI + IIZ)$, we start from the nine terms of $O \otimes O$, back-propagate Pauli-string *pairs* through the reversed circuit, delete locally incompatible trainable pairings via the torus twirl, accumulate input-frequency labels at the encoders, and aggregate the survivors into the exact second moment $S_{\omega\mu} = \mathbb{E}_\theta[a_\omega(\theta)\overline{a_\mu(\theta)}]$. The covariance is then $\Sigma = S - mm^\dagger$ for $m_\omega = \mathbb{E}_\theta[a_\omega(\theta)]$. Every intermediate dictionary size is tracked and matched against the code's own log, and the final analytic covariance is validated against an independent Monte Carlo estimate.

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1 Setup and conventions

1.1 Terminal expectation

Pauli strings are written in qubit order $q_0q_1q_2$, so $ZII = Z_0$, $IZI = Z_1$, and $IIZ = Z_2$. For the worked example we take $\rho = |000\rangle\langle 000|$. This choice only affects the final terminal contraction. During Heisenberg propagation, the observable generally becomes a linear combination of many Pauli strings, including strings with X and Y factors:

$$U(x, \theta)^\dagger O U(x, \theta) = \sum_P c_P(x, \theta) P.$$

These X/Y -containing strings contribute as intermediate components: later gates may rotate or entangle them back into Z -type strings. However, when the fully propagated operator is finally evaluated against $|000\rangle$, only its Z -type Pauli components have non-zero matrix element for this particular input state:

$$\langle 000| P |000\rangle \neq 0 \iff P \in \{I, Z\}^{\otimes 3}.$$

Thus the terminal expectation projects the final Pauli expansion onto the $\{I, Z\}^{\otimes 3}$ components.

1.2 Circuit layout

In the example that we will follow, the ansatz consists of $L = 2$ layers, each of the schematic form

$$\text{RX encoders} \longrightarrow \text{YZY trainable rotations} \longrightarrow \text{triangular CNOT entangler}.$$

Per qubit q the trainable block is $R_{Y_q}(\theta) R_{Z_q}(\theta) R_{Y_q}(\theta)$, and the forward entangler is $\text{CNOT}_{0 \rightarrow 1}$, $\text{CNOT}_{0 \rightarrow 2}$, $\text{CNOT}_{1 \rightarrow 2}$. With three qubits, three trainable rotations per qubit, and two layers, we have $m = 18$ trainable parameters. Figure 1 shows a single layer.

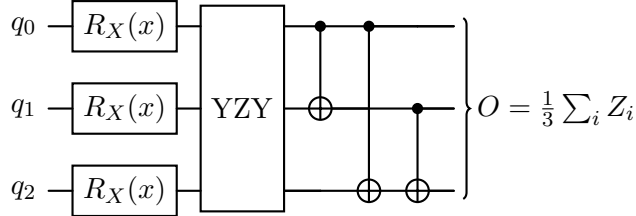


Figure 1: One layer of the ansatz (in our example we repeat this $L = 2$ times). **RX encoders** carry the data x and are *not* averaged; **YZY** gates are trainable and *are* averaged (twirled over $\mathbb{T}^m \cong U(1)^m$); CNOTs are Clifford. The pair-propagation algorithm starts from O on the right and moves left.

Because the forward gate order (within a layer) is encoder, trainable, entangler, the *reversed* order the algorithm processes is entangler, trainable, encoder. Across both layers the reversed sequence of blocks is

$$C_1^{\text{rev}}, T_1^{\text{rev}}, E_1^{\text{rev}}, C_0^{\text{rev}}, T_0^{\text{rev}}, E_0^{\text{rev}}, \quad (1)$$

where C, T, E denote CNOT, trainable, and encoder blocks and the subscript is the layer. The reversed CNOT order within a block is $\text{CNOT}_{1 \rightarrow 2}, \text{CNOT}_{0 \rightarrow 2}, \text{CNOT}_{0 \rightarrow 1}$.

2 Covariance and two-copy of the propagation path

The model is a Fourier series in the data, $f(x, \theta) = \sum_{\omega \in \Omega} a_\omega(\theta) e^{i\omega x}$, whose coefficients are functions of the trainable parameters θ . The object of interest is the coefficient covariance over the parameter ensemble (either the whole torus or otherwise),

$$\Sigma_{\omega\mu} = \text{Cov}_\theta(a_\omega, a_\mu) = S_{\omega\mu} - m_\omega \overline{m_\mu}, \quad S_{\omega\mu} = \mathbb{E}_\theta[a_\omega(\theta) \overline{a_\mu(\theta)}], \quad m_\omega = \mathbb{E}_\theta[a_\omega(\theta)]. \quad (2)$$

Expanding each a_ω over its parameter harmonics gives the identity $\Sigma = CPC^\dagger$, where C is the circuit-harmonic matrix (rows indexed by frequency ω , columns by parameter-harmonic label k) and P projects out the $k = 0$ mode. Materialising C is generally infeasible: the number of active parameter-harmonic columns can grow exponentially.

Second moments from *two* copies of the observable

A mean $m_\omega = \mathbb{E}_\theta[a_\omega]$ is linear in the state and needs *one* copy of the observable. A covariance is a *second* moment $\mathbb{E}_\theta[a_\omega \overline{a_\mu}]$: it is bilinear, so it is the expectation of a two-copy observable $O^{(\omega)} \otimes \overline{O^{(\mu)}}$. Propagating two copies together and averaging each shared trainable parameter over its torus is exactly what $S_{\omega\mu}$, and hence Σ , without forming C . The conjugate on the second copy simply flips the sign of the frequency increments accumulated on that copy; all combinatorics below are otherwise identical on the two copies.

3 The four local gate rules

The whole algorithm is three deterministic Pauli updates plus one averaging rule, applied gate by gate to a dictionary of *pairs* $(Q_1, Q_2, \omega_1, \omega_2) \mapsto \Gamma$ as we propagate backwards.

(1) Clifford / CNOT — deterministic relabelling. Conjugating a Pauli by a Clifford maps it to another Pauli with no branching and no frequency change. For CNOT with control c , target t , the Z -rules are

$$Z_t \mapsto Z_c Z_t, \quad Z_c \mapsto Z_c. \quad (3)$$

(2) Trainable Pauli rotation — twirl, keep/kill/merge. For $R_P(\theta) = e^{-i\theta P/2}$ the single-copy conjugation is $R_P^\dagger Q R_P = Q$ if $[P, Q] = 0$, and $Q \cos \theta + iPQ \sin \theta$ if $\{P, Q\} = 0$. On *two* copies the parameter θ is shared, so the torus average uses the character inner products

$$\langle 1, 1 \rangle = 1, \quad \langle \cos, \cos \rangle = \frac{1}{2}, \quad \langle \sin, \sin \rangle = \frac{1}{2}, \quad \langle 1, \cos \rangle = \langle 1, \sin \rangle = \langle \cos, \sin \rangle = 0. \quad (4)$$

This produces the local pair rule in table 1: a mixed commute/anticommute pair is *killed*; a matched anticommute pair *keeps* its two aligned branches (cos cos and sin sin), each with weight $\frac{1}{2}$.

copy 1	copy 2	two-copy twirl result
commutes	commutes	keep, weight 1
commutes	anticommutes	0 (killed)
anticommutes	commutes	0 (killed)
anticommutes	anticommutes	two matched branches, each weight $\frac{1}{2}$

Table 1: Local rule at a shared trainable gate. The single-qubit letter of P decides commutation: generator X commutes with $\{I, X\}$; Y with $\{I, Y\}$; Z with $\{I, Z\}$.

(3) Merge. Whenever two updated entries share the same tuple of Pauli strings and frequencies, $(Q_1, Q_2, \omega_1, \omega_2)$, their coefficients (as determined by rule (2)), are added.

(4) RX encoder — frequency branching. Encoders are *not* averaged; they create the input-frequency labels. With $R_X(x) = e^{-ixX/2}$,

$$R_X^\dagger Z R_X = Z \cos x + Y \sin x, \quad R_X^\dagger Y R_X = Y \cos x - Z \sin x, \quad R_X^\dagger X R_X = X, \quad (5)$$

and, using $\cos x = \frac{1}{2}(e^{ix} + e^{-ix})$, $\sin x = \frac{1}{2i}(e^{ix} - e^{-ix})$, every Z or Y letter branches as $\omega \mapsto \omega \pm 1$. Unlike parameterised gate branching, these branches are always *retained* because x is left open. This results in a large number of path branching as none are being killed off.

4 The algorithm in pseudocode

Algorithm 1: Two-copy parameter-twirled Pauli propagation for $S_{\omega\mu}$.

Input: observable $O = \sum_s c_s P_s$; reversed gate list $\mathcal{G}$; state $|000\rangle$
Output: second-moment matrix $S_{\omega\mu}$

```

1  $D \leftarrow \{(P_s, P_{s'}, 0, 0) \mapsto c_s \overline{c_{s'}} : s, s'\}$  // two-copy dictionary from  $O \otimes \overline{O}$ 
2 foreach gate  $g$  in  $\mathcal{G}$  do
3   if  $g$  is Clifford then
4     relabel  $Q_1, Q_2$  deterministically on every entry // no branching
5   else if  $g = R_P(\theta)$  is trainable then
6     foreach entry  $(Q_1, Q_2, \omega_1, \omega_2) \mapsto \Gamma$  do
7       if one side commutes, other anticommutes then drop entry
8       else if both commute then keep unchanged
9       else
10        replace by  $\frac{1}{2}(Q_1, Q_2) + \frac{1}{2}(Q'_1, Q'_2)$ , the aligned cos cos, sin sin branches
11   else //  $g = R_X(x)$  encoder
12     branch each  $Z/Y$  letter into  $\omega \mapsto \omega \pm 1$  with cos / sin Fourier weights
13     (on copy 2 the increments carry the opposite sign, from the conjugate)
14   merge entries sharing  $(Q_1, Q_2, \omega_1, \omega_2)$  // add coefficients
15 foreach surviving  $(Q_1, Q_2, \omega_1, \omega_2) \mapsto \Gamma$  do
16    $S_{\omega_1\omega_2} += \Gamma \langle 000 | Q_1 | 000 \rangle \langle 000 | Q_2 | 000 \rangle$  // nonzero only for  $I/Z$  strings
17 return  $S$ 

```

5 Worked example

5.1 Initial two-copy dictionary

The observable is $O = \frac{1}{3}(ZII + IZI + IIZ)$, so the two-copy object starts from the $3 \times 3 = 9$ Pauli-string pairs of $O \otimes O$, each with coefficient $\frac{1}{9}$ and labels $(\omega_1, \omega_2) = (0, 0)$ (fig. 2).

$O \otimes O$: nine pairs, each $\Gamma = \frac{1}{9}$

(ZII, ZII)	(ZII, IZI)	(ZII, IIZ)
(IZI, ZII)	(IZI, IZI)	(IZI, IIZ)
(IIZ, ZII)	(IIZ, IZI)	(IIZ, IIZ)

$O = \frac{1}{3}(ZII + IZI + IIZ)$

Figure 2: The second moment starts from all $3^2 = 9$ observable-term pairs.

5.2 First reversed CNOT entangler

Reiterated again for clarity, in the Heisenberg picture the CNOTs are processed in reverse circuit order, but their control–target orientations are unchanged. For a CNOT with control c and target t , the relevant Z -type propagation rules are

$$Z_c \mapsto Z_c, \quad Z_t \mapsto Z_c Z_t.$$

Thus a Z on the target pulls back an additional Z onto the control, while a Z on the control is unchanged. If both are present, the duplicate control factor cancels, since $Z_c^2 = I$; for example,

$$Z_c Z_t \mapsto Z_c(Z_c Z_t) = Z_t.$$

The reversed CNOTs therefore act deterministically on the one-copy support $\{ZII, IZI, IIZ\}$:

$$\text{CNOT}_{1 \rightarrow 2} : \{ZII, IZI, IIZ\} \mapsto \{ZII, IZI, IZZ\}, \quad (6)$$

$$\text{CNOT}_{0 \rightarrow 2} : \{ZII, IZI, IZZ\} \mapsto \{ZII, IZI, ZZZ\}, \quad (7)$$

$$\text{CNOT}_{0 \rightarrow 1} : \{ZII, IZI, ZZZ\} \mapsto \{ZII, ZZI, IZZ\}. \quad (8)$$

The three observable terms have therefore become

$$A = ZII, \quad B = ZZI, \quad C = IZZ.$$

Since Clifford conjugation maps each Pauli string to a single Pauli string, no terms are created or deleted at this stage. The two-copy dictionary is therefore still the 3×3 set of Cartesian pairs formed from $\{A, B, C\}$, with all entries still carrying frequency label $(0, 0)$.

5.3 First trainable gate $R_{Y_2}(\theta_{17})$

Against generator Y_2 : $A = ZII$ (qubit-2 letter I) commutes, $B = ZZI$ (I) commutes, $C = IZZ$ (Z) anti-commutes. Applying table 1 gives fig. 3: the four mixed $(*, C)$ and $(C, *)$ pairs vanish; the matched (C, C) pair splits into an unchanged C cosine branch and the updated sine branch with C' such that $\frac{1}{2}(C, C) + \frac{1}{2}(C', C')$ with $C' = IZX$. The coefficients of $\frac{1}{2}$ coming from the $\langle \cos, \cos \rangle = \langle \sin, \sin \rangle = \frac{1}{2}$ after twirling.

copy1\copy2	A	B	C
A	keep	keep	kill
B	keep	keep	kill
C	kill	kill	$\frac{1}{2}(C, C) + \frac{1}{2}(C', C')$

Figure 3: The first non-trivial pair twirl, with $A = ZII$ and $B = ZZI$ commuting with Y_2 , $C = IZZ$ anticommutes, and $C' = IZX$ the sin branch. Incompatible pairings are deleted *at this gate*, not at the end.

Since every pair carried weight $\frac{1}{9}$, the dictionary after $R_{Y_2}(\theta_{17})$ is

Q_1	Q_2	Γ
ZII	ZII	$1/9$
ZII	ZZI	$1/9$
ZZI	ZII	$1/9$
ZZI	ZZI	$1/9$
IZZ	IZZ	$1/18$
IZX	IZX	$1/18$

We can immediately see the benefit of twirling as we propagate two copies of the observable. We kill off anti branches that wouldn't survive in the second moment of O as we go, managing the exponential growth of Pauli terms to only those that will actually contribute.

5.4 Remaining qubit-2 gates

Next is $R_{Z_2}(\theta_{16})$. Only IZX anticommutes with Z_2 , so $(IZX, IZX) : \frac{1}{18} \mapsto \frac{1}{36}(IZX, IZX) + \frac{1}{36}(IZY, IZY)$. Then $R_{Y_2}(\theta_{15})$: both IZZ and IZX anticommute with Y_2 , while IZY commutes, and the two anticommute-anticommute entries split and merge,

$$(IZZ, IZZ) : \frac{1}{36} + \frac{1}{72} = \frac{1}{24}, \quad (IZX, IZX) : \frac{1}{36} + \frac{1}{72} = \frac{1}{24}, \quad (9)$$

with IZY left at $\frac{1}{36}$. The same commute/anticommute bookkeeping is then applied to qubits 1 and 0 of the same YZY block; after the full final-layer trainable block the dictionary holds 21 pair states, all still at $(\omega_1, \omega_2) = (0, 0)$ because no encoder has yet been crossed.

5.5 RX encoder: appearance of ω

Crossing $R_{X_0}(x)$ sends $ZII \mapsto ZII \cos x + YII \sin x$ on each copy, so a single pair $(ZII, ZII, 0, 0)$ produces frequency-labelled children (fig. 4). Repeating across the three encoders of the layer drives the jump in number of Pauli string pairs from $21 \rightarrow 63 \rightarrow 191 \rightarrow 291$.

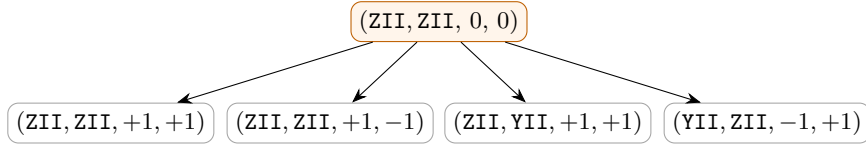


Figure 4: An encoder creates the ω labels that become the entries of the coefficient covariance matrix. Under $R_{X_0}(x)$ on both copies, each Z/Y letter branches $\omega \mapsto \omega \pm 1$ with $\cos/\sin$ weights; unlike a trainable gate this is *not* twirled away.

5.6 Full state-count trace

Table 2 and fig. 5 record the dictionary size through all 30 reversed operations. Trainable twirl steps can kill or merge states; encoder steps create frequency-labelled branches and grow the support.

#	reversed op	type	size	#	reversed op	type	size
0	$O \otimes O$	init	9	16	$\text{CNOT}_{1 \rightarrow 2}$	Clifford	291
1	$\text{CNOT}_{1 \rightarrow 2}$	Clifford	9	17	$\text{CNOT}_{0 \rightarrow 2}$	Clifford	291
2	$\text{CNOT}_{0 \rightarrow 2}$	Clifford	9	18	$\text{CNOT}_{0 \rightarrow 1}$	Clifford	291
3	$\text{CNOT}_{0 \rightarrow 1}$	Clifford	9	19	$R_{Y_2}(\theta_8)$	twirl	232
4	$R_{Y_2}(\theta_{17})$	twirl	6	20	$R_{Z_2}(\theta_7)$	twirl	285
5	$R_{Z_2}(\theta_{16})$	twirl	7	21	$R_{Y_2}(\theta_6)$	twirl	285
6	$R_{Y_2}(\theta_{15})$	twirl	7	22	$R_{Y_1}(\theta_5)$	twirl	312
7	$R_{Y_1}(\theta_{14})$	twirl	9	23	$R_{Z_1}(\theta_4)$	twirl	464
8	$R_{Z_1}(\theta_{13})$	twirl	13	24	$R_{Y_1}(\theta_3)$	twirl	537
9	$R_{Y_1}(\theta_{12})$	twirl	13	25	$R_{Y_0}(\theta_2)$	twirl	536
10	$R_{Y_0}(\theta_{11})$	twirl	17	26	$R_{Z_0}(\theta_1)$	twirl	672
11	$R_{Z_0}(\theta_{10})$	twirl	21	27	$R_{Y_0}(\theta_0)$	twirl	672
12	$R_{Y_0}(\theta_9)$	twirl	21	28	$R_{X_2}(x)$	encoder	1698
13	$R_{X_2}(x)$	encoder	63	29	$R_{X_1}(x)$	encoder	4056
14	$R_{X_1}(x)$	encoder	191	30	$R_{X_0}(x)$	encoder	8402
15	$R_{X_0}(x)$	encoder	291				

Table 2: Pair-dictionary size through the 30 reversed operations ($n = 3, L = 2, d = 1$).

5.7 Terminal evaluation

After all 30 operations each survivor has the form $(Q_1, Q_2, \omega_1, \omega_2) \mapsto \Gamma$, and contributes

$$S_{\omega_1 \omega_2} += \Gamma \langle 000 | Q_1 | 000 \rangle \langle 000 | Q_2 | 000 \rangle. \quad (10)$$

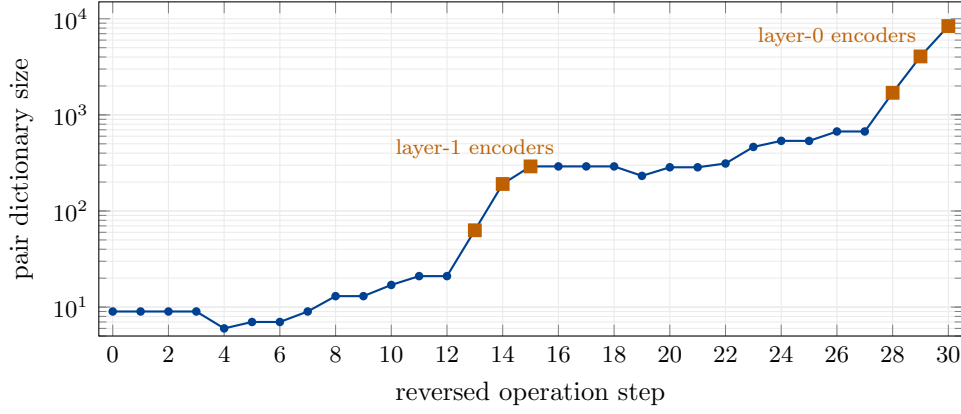


Figure 5: Growth of the pair dictionary. Trainable steps stay flat, shrink or grow slightly (for our particular ansatz); the **encoder steps** (squares) produce rapid growth in the number of pair contributions.

On $|000\rangle$ only $\{I, Z\}$ -strings survive, so any entry carrying an X or Y on either copy drops out at this final step. For the traceless observable used here (mean magnetisation) and uniform YZY parameter average, the one-copy mean vanishes: every non-identity Pauli component is killed by at least one local trainable twirl, and O has no identity component. Thus $m_\omega = 0$ for all ω , so the covariance is simply just the second moment, $\Sigma_{\omega\mu} = S_{\omega\mu}$.

6 Validation against Monte Carlo

Figure 6 compares the exact analytic two-copy twirl with an independent Monte Carlo estimate of the coefficient statistics for this setup. The relative-to-maximum variance profile is essentially identical (Pearson 1.000), and the absolute correlation matrix reproduces the same structured, near-diagonal support. The plot reports support size 11/13 (which means $\omega_{\max} = \pm 5$), Frobenius error $5.213 \times 10^{-3} \pm 8.1 \times 10^{-4}$, off-diagonal cosine similarity 0.998 ± 0.001 , and mean off-diagonal magnitude 8.997×10^{-3} (analytical) versus $9.956 \times 10^{-3} \pm 2.6 \times 10^{-4}$ (MC).

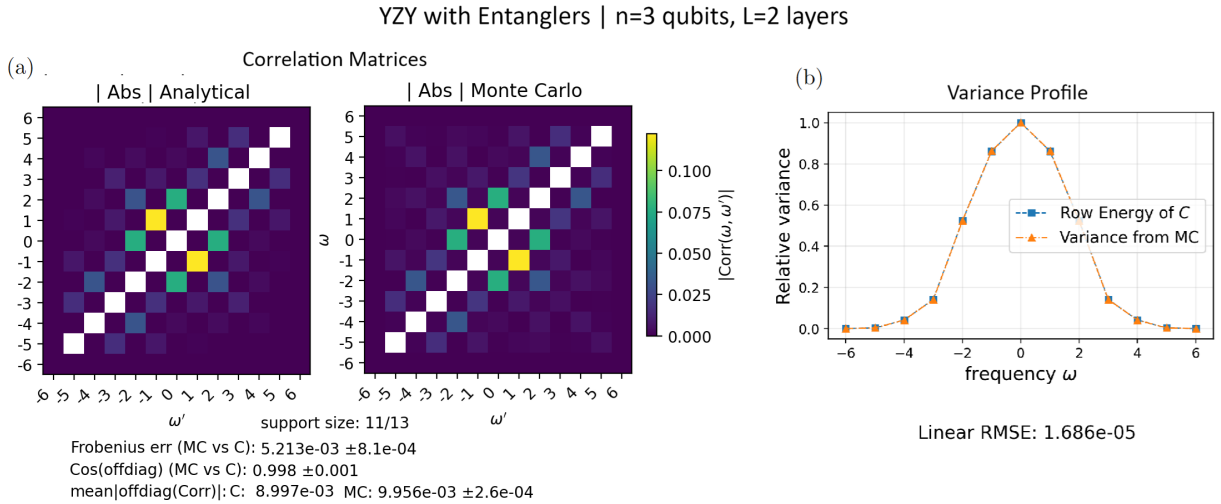


Figure 6: (a) Absolute correlation matrix: exact twirl (left) vs Monte Carlo (right). (b) Relative-to-maximum variance profile. Code output for the worked run. The exact two-copy twirl and Monte Carlo agree on both the variance profile and the off-diagonal correlation structure.

7 Summary

1. Start from the nine explicit terms of $O \otimes O$.
2. Apply reversed CNOTs deterministically to both copies.
3. At each trainable gate, read the relevant single-qubit letter in Q_1 and Q_2 ; if one side commutes and the other anticommutes, discard the pair *immediately*; if both anticommute, keep only the aligned coscos and sinsin branches (each weight $\frac{1}{2}$).
4. Merge entries with identical $(Q_1, Q_2, \omega_1, \omega_2)$.
5. At encoders, branch into $\omega \mapsto \omega \pm 1$ instead of averaging over x .
6. Evaluate survivors on $|000\rangle$ and accumulate into $S_{\omega\mu}$.

The most instructive local step representative of the algorithm is the first trainable gate,

$$\{\text{ZII}, \text{ZZI}, \text{IZZ}\}^{\otimes 2} \xrightarrow{R_{Y_2}(\theta_{17}) \text{ twirl}} \{(\text{ZII}, \text{ZII}), (\text{ZII}, \text{ZZI}), (\text{ZZI}, \text{ZII}), (\text{ZZI}, \text{ZZI}), (\text{IZZ}, \text{IZZ}), (\text{IZX}, \text{IZX})\},$$

where the four mixed pairings vanish because one side carries a constant factor while the other carries $\cos \theta_{17}$ or $\sin \theta_{17}$, and those cross terms integrate to zero over the torus. By twirling in this way, we can compute $CPC^\dagger$ directly without exponentially exploding Pauli propagation terms: the surviving matched branches are exactly the $\sum_{k \neq 0} C_{\omega k} \overline{C_{\mu k}}$ terms by construction, so this acts as both a useful tool in employing C matrices for statistical analysis, but also verifying that C matrices do indeed represent an aggregation of Pauli string coefficients that represent the particular choice of ansatz, observable, input state, and data encoding method.

Related tools/concepts. The construction combines three ideas. Heisenberg-picture Pauli propagation [1, 2, 3] evolves the observable backwards as a sparse sum of Pauli strings. The Fourier structure of data-encoded models [4, 5] means RX encoders generate the integer frequency labels ω . And a covariance is a second moment, so it is naturally represented on two copies and of the back propagated observable averaged by a *twirl* [6, 7]; (note that usually twirling involves the Haar unitary group), on the parameter torus $\mathbb{T}^m \cong \text{U}(1)^m$, and character orthogonality on that torus produces the simple twirling rules of table 1.

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